

CR-004: Student Enrollment Summary and Sorting

1. Change Request Details

Change Request ID: CR-004

Title: Student Enrollment Summary and Sorting

Priority: Low

Difficulty: Easy

Change Type: Frontend Change

2. Business Requirement

Students must be able to view a clear summary of their enrolled courses and the total value of those courses.

The My Enrollments page must provide useful summary information in addition to the individual enrollment cards. Students must also be able to sort their enrolled courses according to enrollment date, price, and course title.

The same total enrolled-course value must also be displayed on the Student Dashboard to provide consistent information across the student-facing pages.

The implementation must use the existing enrollment API response and must not require an additional API request.

3. Current Behaviour

- The My Enrollments page displays individual enrollment cards
- The page does not display the total number of enrolled courses.
- The page does not display the total value of enrolled courses.
- The page does not display the average course price.
- The page does not display the number of distinct course categories.
- There are currently no sorting options for enrolled courses.
- The default enrollment order is not explicitly defined as newest first.
- The original enrollment array may potentially be modified when sorting.
- Course prices may be returned as strings or numeric values.
- Invalid or unsafe price conversion could potentially result in NaN.
- The Student Dashboard does not display the same total enrolled-course value.
- Empty enrollment data does not have a dedicated enrollment empty state.

4. Requested Change

Improve the student enrollment functionality by adding summary information and sorting options to the My Enrollments page.

The My Enrollments page must provide:

- Total number of enrolled courses
- Total value of enrolled courses
- Average course price
- Number of distinct categories
- Sorting by enrollment date, price, and course title
- Newest enrolled as the default sorting option
- Safe handling of course price values
- Protection against NaN values
- A suitable empty state when no courses are enrolled

The Student Dashboard must also display the same total enrolled-course value.

The summary information must be calculated from the existing API response. No additional API request is required.

5. Scope

This change applies to the student enrollment and student dashboard frontend functionality.

The change includes:

- Enrollment summary information
- Total enrolled-course count
- Total enrolled-course value
- Average course price
- Distinct category count
- Enrollment sorting options
- Default newest-first sorting
- Safe price conversion
- Protection against NaN display values
- Student enrollment empty state
- Display of total enrolled-course value on the Student Dashboard
- Reuse of the existing enrollment API response

The change does not include:

- Backend API changes
- Database schema changes
- Additional API requests
- Changes to enrollment data storage
- Changes to enrollment creation or deletion functionality

6. Functional Requirements

6.1 Enrollment Summary Requirements

FR-001: Total Enrolled Courses

The My Enrollments page must display the total number of courses currently enrolled by the student.

For example:

```
Total Enrolled Courses: 5
```

The value must be calculated from the existing enrollment API response.

FR-002: Total Enrolled Course Value

The My Enrollments page must display the total value of all enrolled courses.

The total must be calculated by summing the prices of the enrolled courses.

For example, if the enrolled course prices are:

```
100
```

```
250
```

```
150
```

the displayed total must be:

```
500
```

FR-003: Average Course Price

The My Enrollments page must display the average price of the student's enrolled courses.

The average must be calculated from the enrolled course prices.

If the student has no enrollments, the page must not attempt to display an invalid average.

FR-004: Distinct Category Count

The My Enrollments page must display the number of distinct categories represented by the student's enrolled courses.

For example, if the enrolled courses belong to:

```
Programming
```

```
Programming
```

```
Design
```

```
Business
```

the number of distinct categories must be:

```
3
```

FR-005: Summary Source

All enrollment summary values must be calculated from the existing enrollment API response.

No additional API request may be introduced solely to calculate the summary information.

6.2 Price Handling Requirements

FR-006: Safe Price Conversion

Course prices must be safely converted to numeric values before performing calculations.

The implementation must correctly handle:

- Numeric price values
- Numeric string values
- Zero values

For example:

100

"100"

0

"0"

must be handled correctly.

FR-007: Numeric Calculations

The total value and average price must be calculated using numeric values rather than string concatenation.

For example:

100 + 50

must result in:

150

and not:

10050

FR-008: Prevent NaN

The page must never display NaN as a summary value.

Any invalid or unavailable price value must be handled safely according to the existing application data conventions.

FR-009: Zero Price Handling

A course price of 0 must be treated as a valid numeric price.

A zero-priced course must not be treated as missing simply because its value is falsy.

6.3 Sorting Requirements

FR-010: Sorting Options

The My Enrollments page must provide the following sorting options:

- Newest Enrolled
- Oldest Enrolled
- Price: High to Low
- Price: Low to High
- Course Title: A to Z

FR-011: Default Sorting

The default sorting option must be:

`Newest Enrolled`

When the My Enrollments page is initially loaded, the newest enrollment must appear first.

FR-012: Newest Enrolled Sorting

Selecting `Newest Enrolled` must display the most recently enrolled course first.

FR-013: Oldest Enrolled Sorting

Selecting `Oldest Enrolled` must display the earliest enrolled course first.

FR-014: Price High to Low

Selecting `Price: High to Low` must display the highest-priced course first.

Prices must be compared numerically.

FR-015: Price Low to High

Selecting `Price: Low to High` must display the lowest-priced course first.

Prices must be compared numerically.

FR-016: Course Title Sorting

Selecting `Course Title: A to Z` must sort enrolled courses alphabetically by course title.

Text comparison must be performed consistently and should be case-insensitive.

FR-017: Original Array Protection

Sorting must not modify the original enrollment array received from the API.

The implementation must create an appropriate copy or otherwise use a non-mutating sorting approach before applying sorting.

FR-018: Sorting State

The selected sorting option must remain active while the student remains on the My Enrollments page.

6.4 Empty Enrollment Requirements

FR-019: Empty Enrollment State

If the student has no enrolled courses, the My Enrollments page must display a suitable empty state.

For example:

```
You are not enrolled in any courses yet.
```

FR-020: Empty Summary Values

When there are no enrollments, the page must display safe summary values.

The implementation must not display:

```
NaN
```

```
undefined
```

or other invalid values.

FR-021: Empty Sorting State

When there are no enrollments, the sorting controls must not cause errors.

The page may display the default sorting option while showing the empty enrollment state.

6.5 Student Dashboard Requirements

FR-022: Dashboard Total Value

The Student Dashboard must display the total value of the student's enrolled courses.

FR-023: Consistent Total

The total enrolled-course value displayed on the Student Dashboard must match the total displayed on the My Enrollments page for the same enrollment data.

FR-024: Existing API Response

The Student Dashboard must use the existing available enrollment data and must not introduce an unnecessary additional API request solely for calculating the total.

FR-025: Safe Dashboard Value

The Student Dashboard must not display NaN, undefined, or another invalid value when enrollment data is empty or contains zero-priced courses.

6.6 Existing Functionality Requirements

FR-026: Enrollment Display

Existing enrollment cards must continue to display correctly after the summary and sorting functionality is added.

FR-027: Course Information

Adding sorting and summary functionality must not remove or alter the existing course information displayed on enrollment cards.

FR-028: Non-Destructive Sorting

Changing the selected sorting option must only change the displayed order of the enrollment cards.

It must not modify the underlying enrollment data.

7. Security Requirements

SEC-001: Existing Authentication

Existing student authentication requirements must remain unchanged.

SEC-002: Student Data Isolation

A student must only be able to view enrollment information belonging to the authenticated student.

SEC-003: Existing Authorization

The implementation must preserve existing authorization and access-control behaviour for student enrollment and dashboard pages.

SEC-004: Frontend Data Handling

The frontend must only calculate and display summary information from enrollment data already provided to the authenticated student.

No additional unrestricted data access may be introduced.

8. Configuration / Potentially Affected Components

Frontend

Potentially affected areas include:

```
frontend/src/pages/MyEnrollments.jsx
```

```
frontend/src/pages/StudentDashboard.jsx
```

The exact files may vary depending on the existing implementation.

Backend

No backend changes are expected.

Existing enrollment APIs should continue to provide the data required by the frontend.

API

No API endpoint changes are required.

The existing enrollment API response must be used to calculate the required summary information.

Database

No database changes are required.

Existing enrollment and course data must continue to be used.

Expected / Potentially Affected Files

The following files are examples of expected or potentially affected files, not a mandatory list of files that must all be modified.

- frontend/src/pages/MyEnrollments.jsx
- frontend/src/pages/StudentDashboard.jsx
- frontend/src/App.css

Students must perform an impact analysis before implementation.

If additional files are modified, they must be identified and justified in the Implementation Report.

9. Constraints / Assumptions

- Existing student enrollment functionality should continue to work.
- Existing enrollment API responses should continue to be used.
- No backend changes are required.
- No database changes are required.
- No additional API request is required for the enrollment summary.
- The default sorting order must be newest enrolled first.
- Sorting must not mutate the original enrollment array.
- Price values may be returned as either numbers or numeric strings.
- Price conversion must be safe and deterministic.
- The implementation must not display NaN.
- A price of 0 must be treated as a valid price.
- Empty enrollment data must not cause runtime errors.
- The total displayed on the Student Dashboard must match the total displayed on the My Enrollments page.

- Existing enrollment cards must continue to work correctly.
- Sorting must only affect the presentation order and must not modify stored enrollment data.

10. Acceptance Criteria

AC-001: Total Enrollment Count

The My Enrollments page displays the total number of courses enrolled by the student.

AC-002: Total Enrollment Value

The total value displayed equals the sum of the displayed enrolled course prices.

AC-003: Average Course Price

The My Enrollments page displays the correct average price based on the enrolled courses.

AC-004: Distinct Categories

The My Enrollments page displays the correct number of distinct course categories.

AC-005: Newest Sorting

The `Newest Enrolled` sorting option displays the newest enrollment first.

AC-006: Oldest Sorting

The `Oldest Enrolled` sorting option displays the oldest enrollment first.

AC-007: Price High to Low

The `Price: High to Low` sorting option displays the highest-priced course first.

AC-008: Price Low to High

The `Price: Low to High` sorting option displays the lowest-priced course first.

AC-009: Course Title Sorting

The `Course Title: A to Z` sorting option sorts courses alphabetically by course title.

AC-010: Default Sorting

The default enrollment order is `Newest Enrolled`.

AC-011: Original Array Protection

Changing the sorting option does not modify the original enrollment array.

AC-012: Numeric Price Handling

Numeric prices and numeric string prices are converted and processed correctly.

For example:

100

and:

"100"

must both be treated as the numeric value 100.

AC-013: Zero Price Handling

A course with a price of 0 is processed correctly and included in the total calculation.

AC-014: No NaN

The My Enrollments page never displays NaN for the total, average, or other summary values.

AC-015: Empty Enrollment Data

If the student has no enrollments, the page displays a suitable empty state without causing an error.

AC-016: Empty Summary

Empty enrollment data does not produce NaN, undefined, or other invalid summary values.

AC-017: Dashboard Total

The Student Dashboard displays the total value of the student's enrolled courses.

AC-018: Consistent Dashboard and Enrollment Total

The total displayed on the Student Dashboard matches the total displayed on the My Enrollments page for the same enrollment data.

AC-019: Existing Enrollment Cards

Existing enrollment cards continue to display correctly.

AC-020: No Additional API Request

The summary information is calculated from the existing API response without requiring an additional API request.

AC-021: No Backend Changes

The functionality is implemented without requiring backend changes.

AC-022: No Database Changes

The functionality is implemented without requiring database changes.

AC-023: Sorting Does Not Modify Data

Sorting changes only the displayed order and does not modify the underlying enrollment data.

AC-024: CRUD / Existing Functionality

Existing student enrollment functionality continues to work correctly after the summary and sorting features are implemented.

11. Expected Outcome

After implementation:

- Students can see the total number of their enrolled courses.
- Students can see the total value of their enrolled courses.
- Students can see the average course price.
- Students can see the number of distinct course categories.
- Students can sort their enrollments by newest enrolled, oldest enrolled, price, and course title.
- The default enrollment order is newest first.
- Price values are safely converted from strings or numbers before calculations and sorting.
- Zero-priced courses are handled correctly.
- The application never displays `NaN` for enrollment summary values.
- The original enrollment array is not modified during sorting.
- Students receive a suitable empty state when they have no enrollments.
- The Student Dashboard displays the same total enrolled-course value as the My Enrollments page.
- Summary information is calculated using the existing API response.
- No additional API request is required.
- Existing enrollment cards and student functionality continue to work correctly.
- No backend or database changes are required.